



TITLE PAGE

- Problem Statement ID – 26149
- Problem Statement Title - Design and Development of an Integrated Secure Data Erasure and Advanced File Recovery Tool for Digital Forensics and Data Sanitization
- Theme - Blockchain& Cybersecurity
- PS Category- Software
- Team ID -
- Team Name (Registered on portal): Gigabyte



Proposed Solution :

An integrated platform combining secure data sanitization and advanced file recovery with verification and forensic report.

Key Features

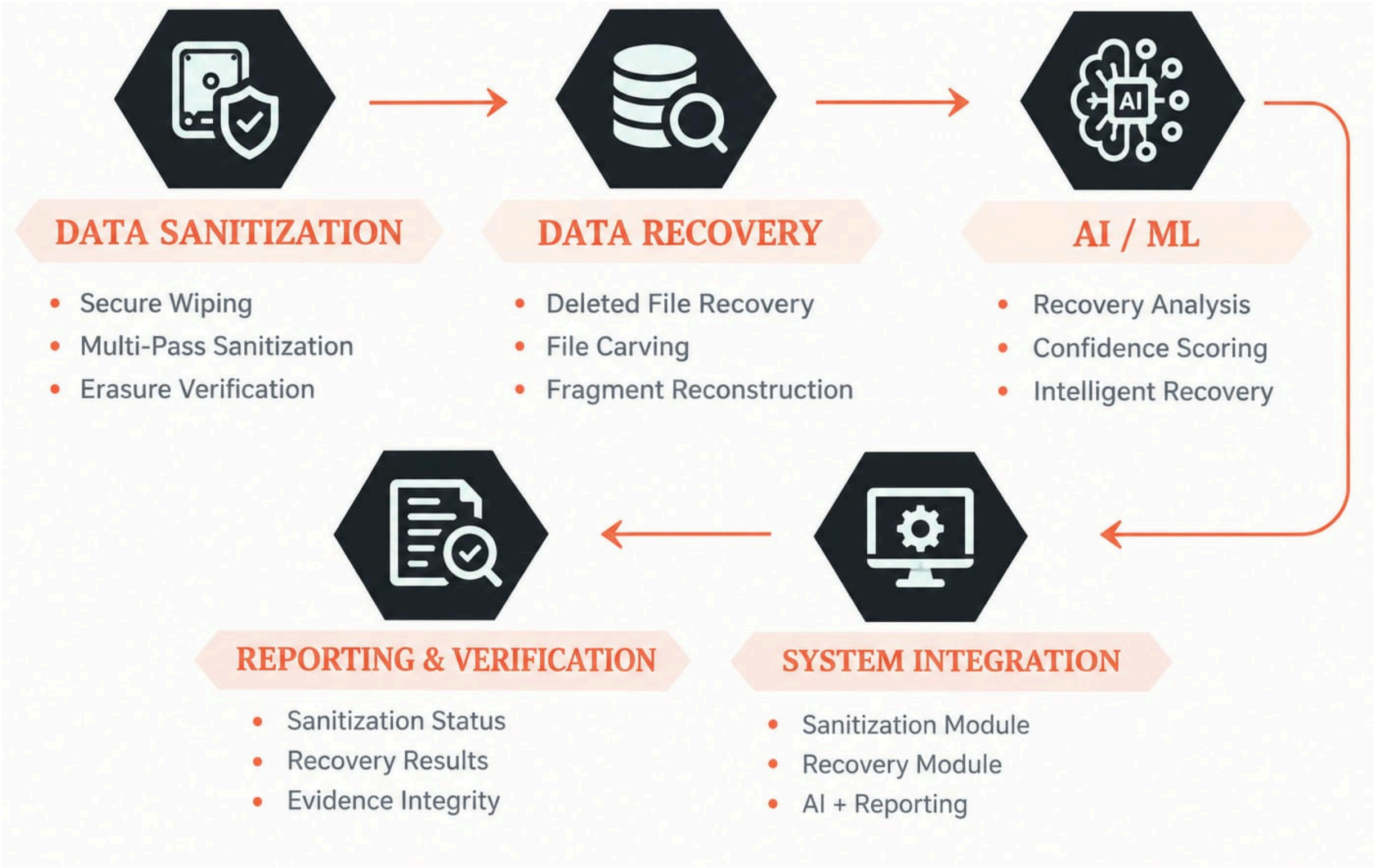
- **Smart & Secure Sanitization** – Securely erases drives, files and folders while recommending suitable sanitization methods.
- **Advanced File Recovery** – Recovers deleted or lost files using file carving.
- **Audit & Forensic Reports** – Maintains logs and generates tamper-evident reports.

Innovation and Uniqueness

- Sanitization + Recovery in one platform
- Intelligent file carving/reconstruction
- Verification + tamper-evident reporting
- Designed as a unified forensic workflow

How It Solves the Problem

- **Tool Fragmentation** → Combines data sanitization and forensic recovery in one unified platform.
- **Data Security** → Securely sanitizes sensitive data with verification.
- **Deleted Evidence** → Recovers deleted files using advanced file carving techniques.





Established Forensic Techniques

Uses proven digital forensic methods.



Sleuth Kit Integration

Enables filesystem-aware file recovery.



Signature-Based Carving

Recovers files when filesystem metadata is unavailable.



Accessible Tools

Can be implemented using widely available software tools.



Controlled Testing

Development and testing can be performed safely using disk images.



Reliable Results

Multiple techniques increase the chances of successful recovery.



Validation

Validates recovered files for improved reliability.



Multiple Recovery Paths

Combines filesystem recovery and raw file carving.



Practical Requirement

Deleted-file recovery is essential in digital forensics.

IMPACT

Trustworthy & Verifiable

Provides verification, encryption and forensic reports for transparent and reliable results.



Secure Data Lifecycle

Ensures sensitive data is securely erased before devices are reused, transferred or disposed.

Faster Investigations

Enables efficient recovery of deleted or fragmented files, supporting digital forensic investigations.

BENEFITS



Data Security

Protects sensitive and confidential data by securely erasing files, folders and storage devices.

Economic

Reduces data-breach risks and enables safe reuse or disposal of storage devices.



Forensic Support

Helps investigators recover deleted evidence and generate reliable forensic information.

- **Academic Research**

1. **Forensic Science International: Digital Investigation (2023)** : Identification of data wiping tools based on deletion patterns in ReFS \$Logfile — research on data wiping tools, deletion patterns, and forensic detection.
2. **arXiv Research (2026)** : Byte-level generative predictions for forensics multimedia carving — explores generative models for fragmented file recovery and byte-level fragment prediction.
3. **CUSTOS** : Toward Forensic-Ready Zero Trust at the Capture-Containment Boundary

- **Technical Resources**

1. **The Sleuth Kit (TSK)** : Sleuth kit Documentation and developers guide.